ILESH PATEL

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Logicrest TECHNETICS IT SERVICES

Professional Summary

Results-driven Information Technology student specializing in **Artificial Intelligence and Machine Learning**, with strong expertise in **Python**, data structures, and database systems. Experienced in data preprocessing, exploratory data analysis, feature engineering, and building **supervised and unsupervised learning models** using Scikit-learn, Pandas, and NumPy. Proficient in **Deep Learning architectures** including CNNs for computer vision and RNN-based models for sequence and time-series data. Skilled in **Natural Language Processing and Generative AI**, including embeddings, prompt engineering, and large language model integration. Focused on designing **end-to-end AI systems** and integrating scalable ML solutions into real-world applications that deliver measurable impact.

Skills

Artificial Intelligence & Machine Learning

- Deep Learning
- Neural Networks, Transformers
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- NLP & Semantic Search
- Embeddings & Vector Similarity
- Prompt Engineering & Evaluation
- Model Benchmarking & Error Analysis
- Hallucination Mitigation & Guardrails
- Feature Engineering & Data Preprocessing

Generative AI Systems

- LLM Application Design
- Context Management & Chunking Strategies
- Multi-step Reasoning Pipelines
- AI Agents & Tool-calling Workflows
- Structured Output Generation

Model Deployment & Inference Engineering

- Deploying ML models using Flask
- Building simple REST APIs for predictions
- Basic understanding of inference workflows and latency

Frameworks & Libraries

- Pandas, NumPy, Scikit-learn
- PyTorch, TensorFlow
- Matplotlib, Seaborn
- Flask / FastAPI
- Hugging Face Transformers

Web Programming (Supporting Skills)

- HTML5, CSS3, JavaScript
- Responsive UI design (Bootstrap/Tailwind)
- Node.js & Express.js (backend development)
- Docker
- REST APIs & Async Services

Database Management Systems

- DBMS/RDBMS fundamentals (normalization, indexing, transactions)
- MySQL (queries, joins, aggregations)
- PostgreSQL (relational database design and integration)
- MongoDB (NoSQL document-based data modeling)

Programming Tools

- Git & GitHub
- VS Code / Jupyter Notebook
- Docker
- MS Excel

Programming Language

- Python (Primary)
- SQL, JavaScript, HTML, CSS (Supporting)

Soft Skills

- Problem-solving and Analytical Thinking
- Attention to Detail
- Flexibility and Active Learning
- Communication & Presentation
- Adaptability & Team Collaboration

Experience

Technetics IT Services – AI/ML Intern *Jan 2026 – Present | Surat, Gujarat, India*

- Built a document question-answering system using **Retrieval-Augmented Generation (RAG)** with embeddings to retrieve and generate context-aware responses from datasets.
- Implemented text preprocessing, chunking, and semantic search using vector embeddings to improve retrieval relevance.
- Developed a **Variational Autoencoder (VAE)** for feature learning and basic anomaly detection, improving data representation quality.
- Evaluated model performance using standard metrics such as **accuracy**, **reconstruction loss**, and **precision@k**.
- Integrated trained ML models into a FastAPI backend to provide prediction APIs for web-based applications.
- Used **Docker** for local deployment and version control to ensure consistent development environments.

Logicrest – Research Intern (Hardware Team Lead) *Sep 2025 – Dec 2025 | Surat, Gujarat, India (Remote)*

- Led the hardware intern team in researching and selecting industrial-grade sensors and devices for implementation in a **Sewage Treatment Plant (STP)** monitoring system.
- Conducted technical analysis of water treatment methodologies including biological, chemical, and filtration-based treatment processes.
- Researched and evaluated water quality sensors (pH, turbidity, TDS, dissolved oxygen, flow sensors) for real-time monitoring applications.
- Studied control units and automation systems including microcontrollers and PLC-based architectures for process regulation.
- Assisted in designing a hardware framework for integrating sensors with monitoring and data collection systems.

Projects

AI-Powered Resume Analyzer & Career Intelligence Platform - Demo

- Built an end-to-end AI system that analyzes resumes to provide **Role matching, ATS scoring, Skill gap Detection, and career path recommendations** using NLP and semantic similarity techniques.
- Designed a hybrid Role-matching engine combining semantic embeddings, skill overlap analysis, experience alignment, and keyword relevance for ranked job fit scoring.
- Implemented an explainable **ATS scoring engine** with category-wise breakdown (skills, experience, formatting) and simulated ATS parsing for keyword coverage and readability analysis.
- Developed **Skill normalization and fuzzy matching pipelines** to detect missing, matched, and trending skills based on industry demand.
- Created an interactive **Multi-tab dashboard** with responsive UI to visualize insights including radar charts, role scores, and priority-ranked improvement suggestions.
- Enabled full report generation for structured resume analysis output and actionable career intelligence insights.

AI-Powered Logistics Intelligence & Command System - Demo

- Designed and developed **A human-in-the-loop AI-powered logistics decision support system** to assess pre-dispatch shipment risk using multi-factor analysis.
- Optimized **Risk scoring logic** to improve shipment feasibility assessment consistency and decision clarity.

- Built a hybrid **Rule-based + Machine learning model** (Decision Tree Classifier) to evaluate route complexity, weather impact, vehicle feasibility, and urgency classification.
- Created role-based dashboards (Seller, Manager, Supervisor) using Streamlit with audit logging and override tracking for governance compliance.
- Implemented real-time **Weather risk integration** using external APIs and structured shipment validation pipelines.
- Developed an **Explainable AI workflow** providing plain-English risk explanations to support transparent managerial decision-making.
- Engineered a complete **Decision audit trail system** to ensure traceability, accountability, and operational transparency.

Stock Trend Prediction - Demo

- Developed an AI-based stock forecasting web application using **LSTM neural networks** to predict price trends from historical time-series data with adaptive model selection.
- Engineered a dynamic fallback trend analysis system using moving averages and momentum indicators when **Deep learning** predictions were unavailable.
- Integrated real-time financial data and company metrics using **Yahoo Finance APIs**, enabling live market analysis and interactive candlestick visualizations with Plotly.
- Designed a responsive **Streamlit dashboard** with technical indicators, prediction overlays, and multi-timeframe forecasting (1–90 days) for comprehensive market insights.
- Implemented data preprocessing pipelines including scaling, sequence generation, and volatility analysis using Pandas and Scikit-learn to improve Model generalization.

Education

Bachelor of Engineering

C.K. Pithawalla College of Engineering & Technology – Surat, Gujarat, India

Higher Secondary Certificate (HSC) in Science Stream

Ashadeep IIT – Surat, Gujarat [2023]

Secondary School Certificate (SSC)

K. & M. P. Sarvajanik Vidhyalay – Surat, Gujarat [2021]

Certifications & Achievements

- IBM - Python for Data Science, AI & Development - Credential
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional - Credential
- Google - Crash Course on Python - Credential
- Google - Regression Analysis: Simplify Complex Data Relationships - Credential
- Lead Generation Messenger Chatbot using Chatfuel - Credential

Interests & Activities

- Learning about Generative AI and emerging ML trends through articles, tutorials, and research papers.
- Participating in Technical discussions and sharing knowledge with peers.
- Studying automation and control systems used in smart infrastructure and industrial operations.
- Experimenting with Automation tools and improving workflow efficiency